MACHINE LEARNING

Which Regularizer Should You Actually Use? Lessons fro Simulations

A practitioner's decision framework for Ridge, Lasso based on three quantities you can compute before

Ahsaas Bajaj
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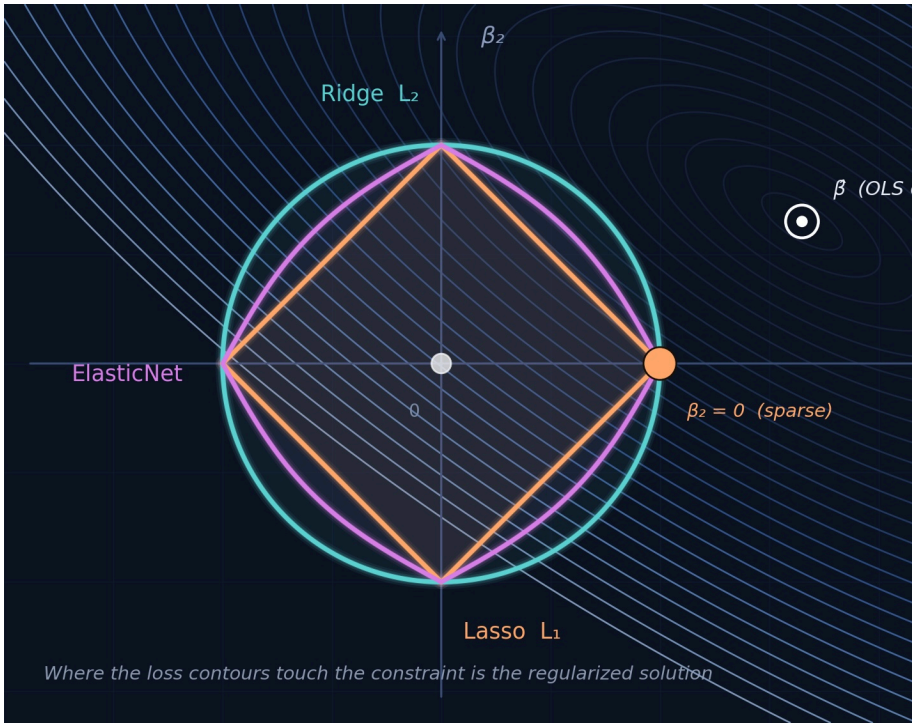


Authors: Ahsaas Bajaj and Benjamin S Knight

Ridge, Lasso, or ElasticNet? We ran 134,400 sim in real production ML models to find out. The a hat you're optimizing for, and on a single diag compute before fitting a model.

If you've ever trained a linear model in scikit-learn, you've faced this question: RidgeCV, LassoCV, or ElasticNetCV? Maybe you defaulted to whatever a tutorial recommended. Maybe a colleague had a strong opinion. Maybe you tried all three and picked whichever gave the best cross-validation score.

We wanted to replace intuition with empirical decision-making.

We ran 134,400 simulations across 960 configu dimensional parameter space, varying sample multicollinearity, signal-to-noise ratio, coeffici two more parameters. We benchmarked four r frameworks (Ridge, Lasso, ElasticNet, and Pos across the three objectives:

1. **Predictive accuracy** (test RMSE)
2. **Variable selection** (F1 score for recovering set)
3. **Coefficient estimation** (L2 error vs. true c

Our simulation ranges aren't arbitrary. They're **real-world production ML models** from Instac demand forecasting, conversion prediction, an intelligence. The regimes we tested reflect cor actually encounter in practice.

This post distills the practical guidance from c decision framework you can use on your next Data Scientist or MLE choosing a regularizer, t

The Headlines

efore we get into the details:

- **For prediction, it barely matters.** Ridge, Lasso, and ElasticNet differ by at most 0.3% in median RMSE. No hyperparameter achieves even a small effect size for RMSE differences among them. This only holds with adequate training data (> 78 observations per feature).
- **For variable selection, it matters enormously, especially under multicollinearity**. Lasso's recall collapses to 0.18 under high condition numbers with low sig ElasticNet maintains 0.93.
- At large sample-to-feature ratios (n/p ≥ 78 become **interchangeable**. Use Ridge; it's th
- **Post-Lasso OLS should be avoided when c RMSE.** It's the only method that consisten and it does so on every objective we meas

What We Tested and Why

Our simulation framework varies seven hyper- simultaneously:

Parameter	Values Tested
Sample size (n)	100 · 1,000 · 10,000 · 100
Features (p)	64 · 128
Condition number (κ)	$\sim 10^1$ (well-conditioned) · $\sim$
Signal-to-noise ratio	0.04 · 0.20 · 1.0
Sparsity	0% (dense) · 15% (sparse)
β distribution	Uniform · Normal
Rank ratio	0.5 · 1.0

Table 1: We simulated a hyperparameter space of 960 con

We ran each of the 4 regularization framework yper-parameter configurations, each using 35

a total of 134,400 simulations. For every simulation we logged the test RMSE, F1 score (precision and recall for recovering the true support of β), and coefficient L2 error.

To measure *what drives the differences between methods*, we used **omega-squared (ω²)** from one-way ANOVA, an effect size that tells us what proportion of variance in performance gaps is explained by each parameter. This goes beyond asking "which method wins" to understanding *why* it wins, ar conditions.

Here's what this means in practice: **most of th drive method differences are things you can o fitting a model.** You know *n* and *p*. You can cor condition number κ with numpy.linalg.cond(X). important latent parameter, SNR, has a free di the regularization strength α that LassoCV sele signals low signal; low α signals strong signal. to this.

Finding 1: For Prediction, Just Use

This is the most important finding for the large practitioners.

Ridge, Lasso, and ElasticNet are nearly interch prediction. Across all 33,600 simulations per r median test RMSE differs by at most 0.3%. Ou analysis confirms this: no single hyperparamet small effect size (ω² ≥ 0.01) for RMSE differenc three methods. Every pairwise comparison is r 0.02).

For practitioners who only care about accuracy, the near-equivalence is itself the finding. Regularizer choice matters far less than sample size.

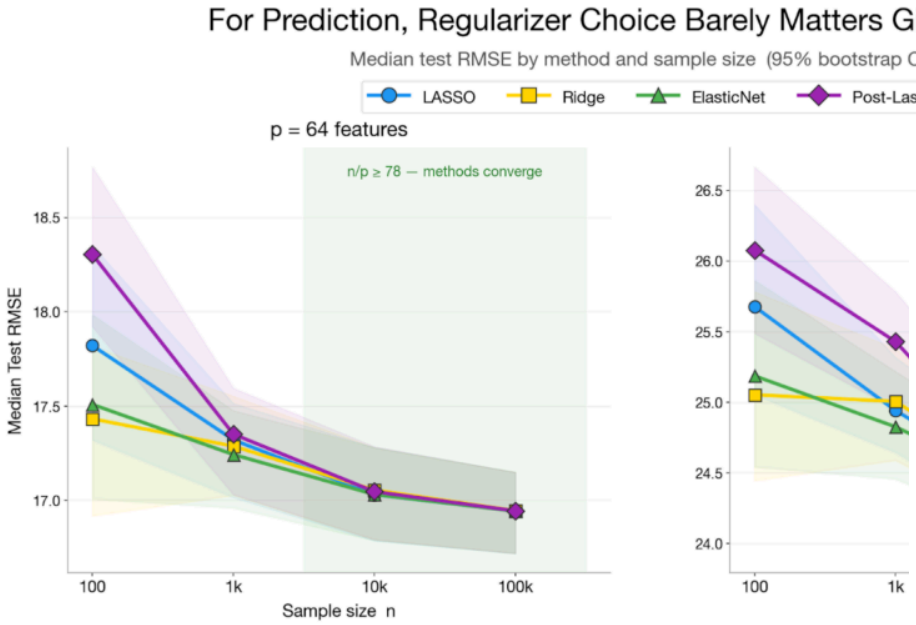


Figure 1: Differences in test RMSE become trivial given sufficie

So why Ridge? Computational efficiency. Ridge form solution for each candidate α, making it than the alternatives (compare Ridge's median seconds to Lasso's median runtime of 9 secon ElasticNet's median runtime of 48 seconds).

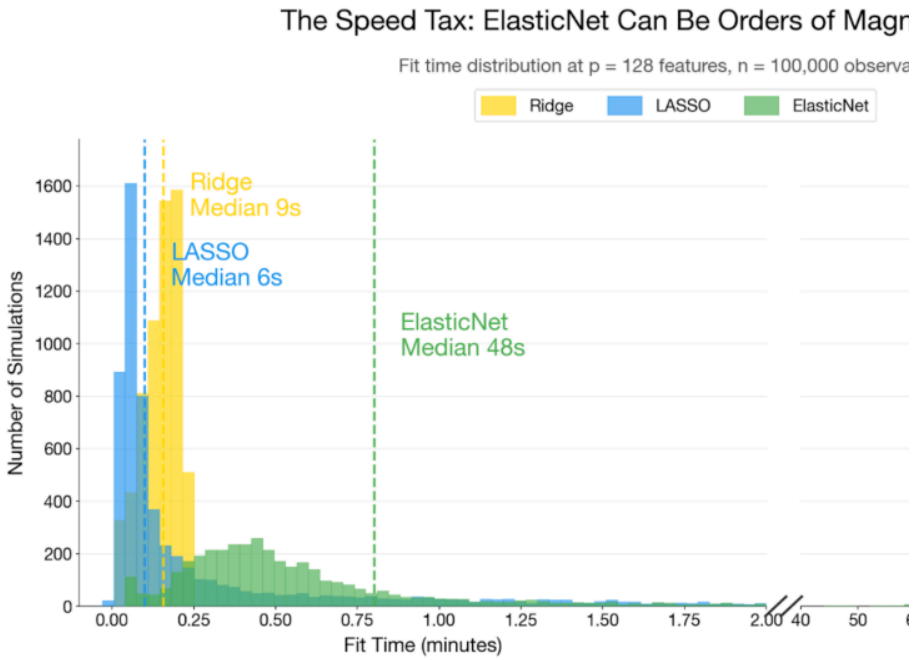


Figure 2: Users should expect a minimum of a 5X increase in runtimes when opting for ElasticNet over Ridge or Lasso.

ElasticNet's overhead stems from its joint grid search over α and the L1 ratio ρ. The 167–219× mean overhead we measured is specific to our 8-value L1 ratio grid. A coarser 3-value grid would reduce this proportionally. Even worse, when the coefficient distribution is approximately uniform, Lasso can take over an hour to converge (see the right-side of the bim This overhead buys you a median RMSE impro 0.04% over Ridge, a margin that's negligible in

Caveats

At the smallest sample size we tested (n = 10(beat Ridge by 5–15% in very specific instances (~1.0). At low SNR, Ridge is actually marginally localized observations at the extreme of our si systematic trends.

One more note: LassoLars wasn't part of our r but the LARS algorithm computes the entire L regularization path analytically in a single pass potentially matching Ridge's closed-form spee However, LARS is known to be numerically uns collinearity conditions ($\kappa > 10^4$) that characteri production ML feature sets. This is precisely th our strongest findings apply.

Bottom line for prediction: Default to RidgeCV. matters far more than regularizer choice. But p only objective worth optimizing. When variable coefficient accuracy matters, especially under ıe story changes dramatically.

Finding 2: For Variable Selection, ElasticNet Is the Safe Default

Here method choice actually matters. Variable selection, the task of identifying which features truly contribute to the outcome, is the objective most sensitive to the regularizer, and where getting it wrong carries the steepest cost.

What Drives the Differences

From our ANOVA decomposition of pairwise F1

Factor	Avg ω^2
Sample size (n)	0.308
SNR *(latent)*	0.065
β distribution *(latent)*	up to 0.112 (ElasticNet vs. Ridge)
Condition number (κ)	0.013

Table 2: Sample size is the most salient predictor of difference

Sample size dominates overwhelmingly. But or small-*n* regime (*n/p* < 78), the condition numb become the primary differentiators.

High Multicollinearity (κ > ~10⁴): Do Not Use La

This is one of the most robust findings in the e it's directly relevant to production ML. **Seven o surveyed operate in the high-κ regime.** If your moderately correlated (which they almost cert engineered feature set), this finding applies to

^t high κ with low SNR:

- **Lasso recall: 0.18** (it misses 82% of true features)
- **ElasticNet recall: 0.93** (it catches 93% of true features)

That's a **5× recall advantage for ElasticNet**. The mechanism is well-known. When features are highly correlated, Lasso arbitrarily picks one from each correlated group and zeros the rest. ElasticNet's L2 penalty component, the "grouping effect" described by Zou and Hastie (2005), keeps cor
together.

Our simulations show this isn't a corner case. differences (ΔF1 of 0.50–0.75) concentrate squ columns at n = 100 and n = 1,000. This is the c production.

Low Multicollinearity (κ < ~10²): Still Default t

You might expect Lasso to finally shine at low least not universally. Even at low κ, Lasso's rec sensitive to the signal-to-noise ratio (see belo

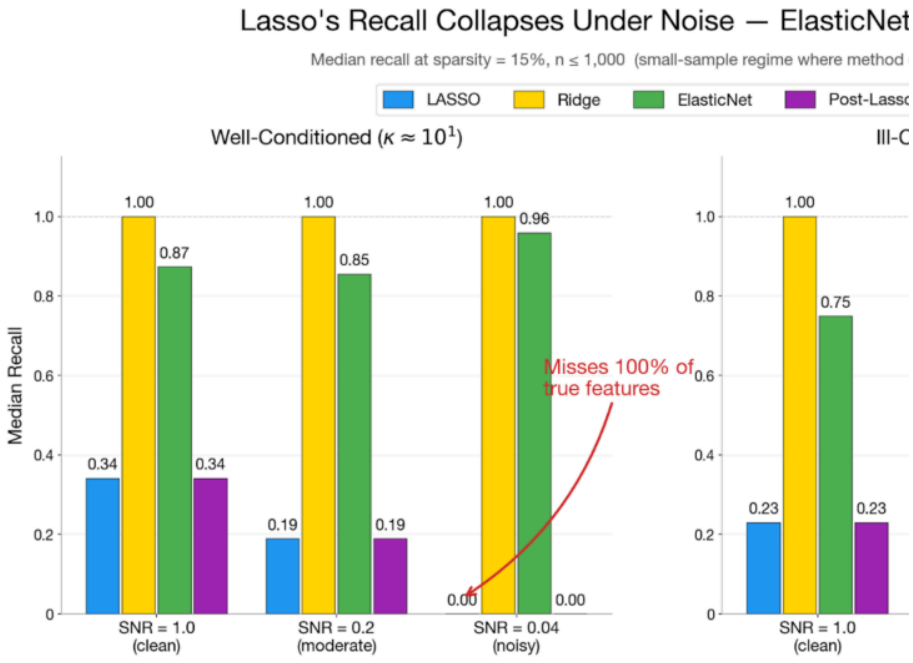


Figure 3: ElasticNet's use of the L2 norm protects against the recall collap

ElasticNet maintains recall ≥ 0.91 regardless of SNR, even at low κ. Lasso is only competitive when *both* SNR is high *and* the true model is genuinely sparse. Since you typically don't know SNR in advance, ElasticNet is the safer bet.

The Ridge Surprise

We didn't expect this: Ridge frequently achieves the *highest* F1 scores at small *n*, despite never performing ex selection. How? Ridge's recall is always 1.0, be every feature, and that perfect recall overwhel advantage of sparse methods when those met collapses under low SNR.

But this isn't genuine variable selection. Ridge nonzero coefficient for every feature. If you ne sparse model, Ridge doesn't help. Combining F hoc permutation importance is a natural exter didn't evaluate it here.

Variable Selection: Summary

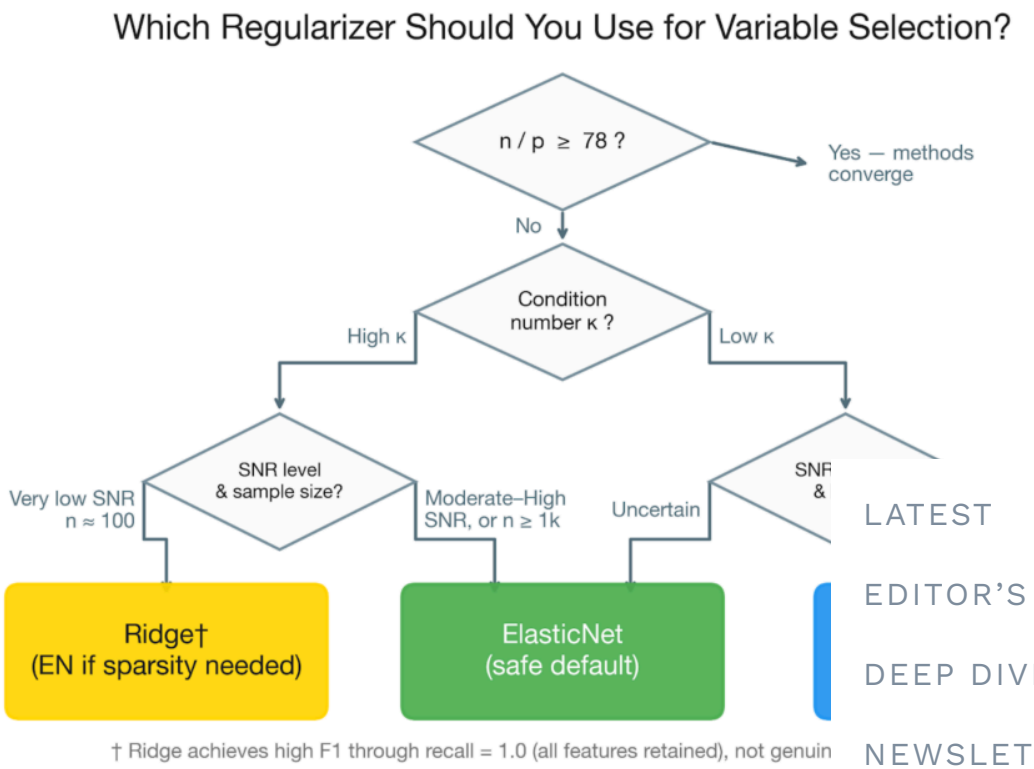


Figure 4: ElasticNet is the safe choice when the researcher canno

Bottom line for variable selection: ElasticNet(default. Lasso only earns its place when κ is lo and you have domain reason to believe the tru

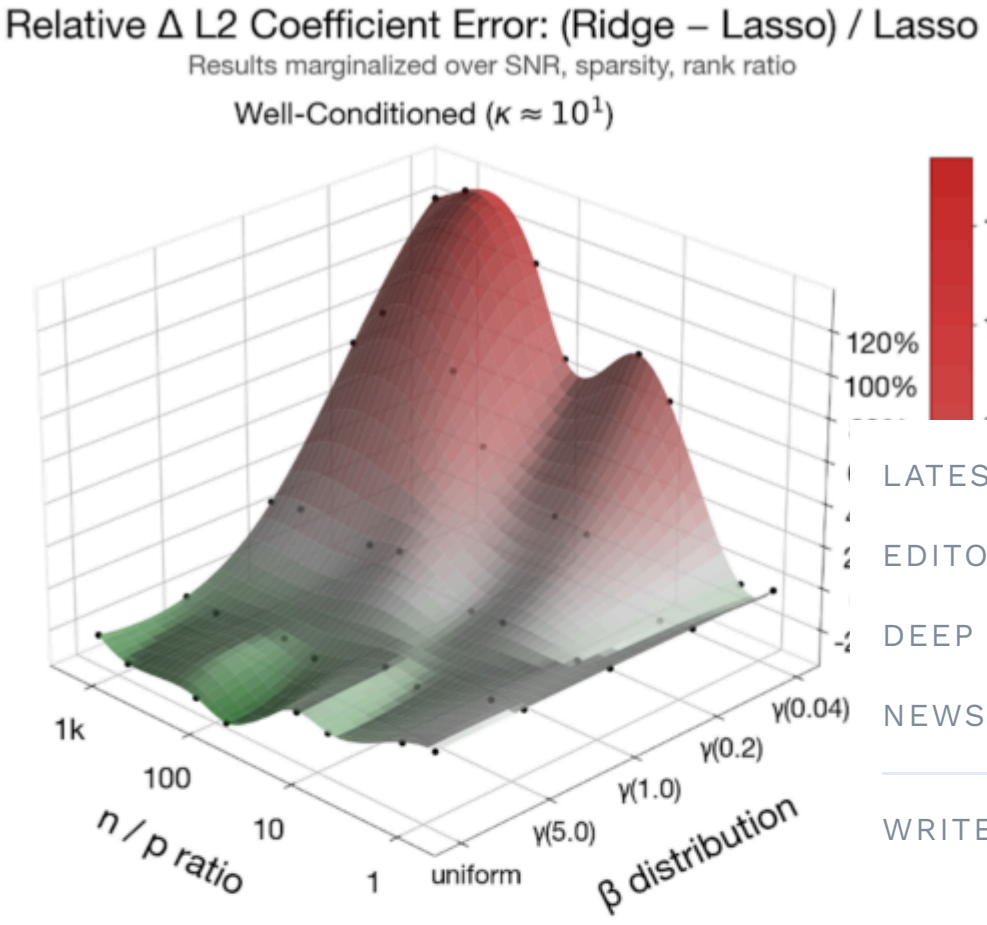
Finding 3: For Coefficient Estimatic κ

When the goal is recovering accurate coefficie interpretability or causal inference, the conditi becomes the key branching variable. Ideally we the distribution of the true β coefficients, but observe it. In contrast, κ can be measured dire ElasticNet dominates regardless of sparsity. At optimal method depends on whether the true dense. Sample size changes the *magnitude* of not their *direction*.

High κ (> ~10^4): Use ElasticNet. It achieves 20–40% lower L2 coefficient error than Lasso, and holds a consistent edge over Ridge regardless of sparsity level.

Low κ (< ~10^2): Branch on your domain knowledge about sparsity.

- *Sparse domain* (genomics, text classification. sensor arravs): Lasso or ElasticNet
- *Dense domain* (engineered feature sets, de conversion models): Ridge



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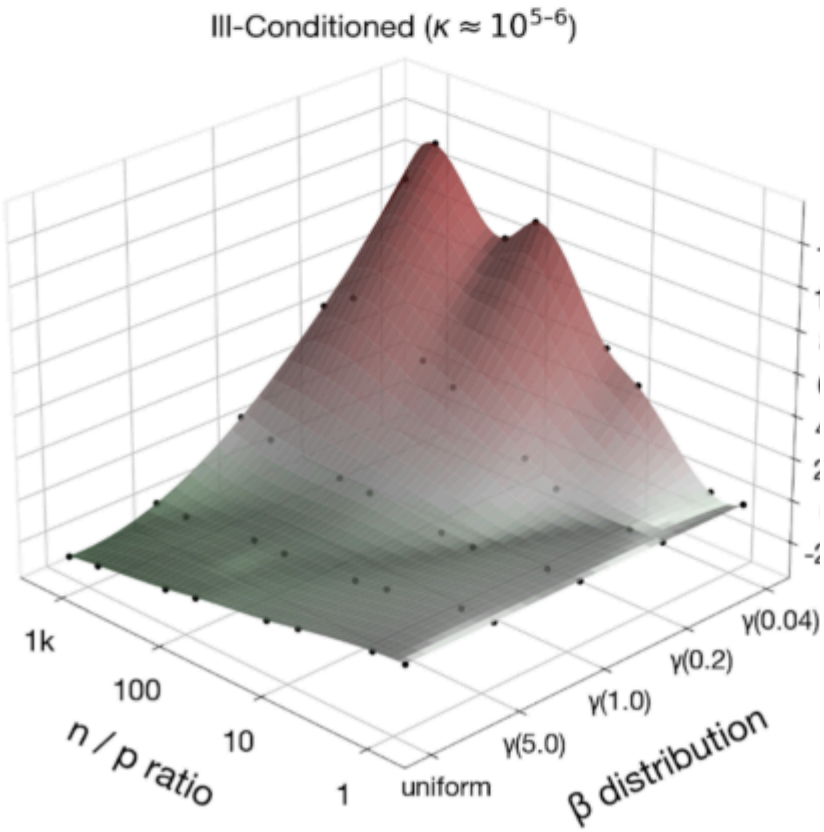

Figure 5: Ridge's performance advantage over Lasso / ElasticNet fades quickly as the n / p ratio increases, while a well-conditioned eigenspace further advantages Lasso / ElasticNet.

All regimes: Avoid Post-Lasso OLS. It shows higher coefficient L2 error than standard Lasso across the entire simulation grid. The unpenalized OLS refit amplifies first-stage selection errors. This is the scenario where you'd hope the two-stage procedure helps, and it doesn't.

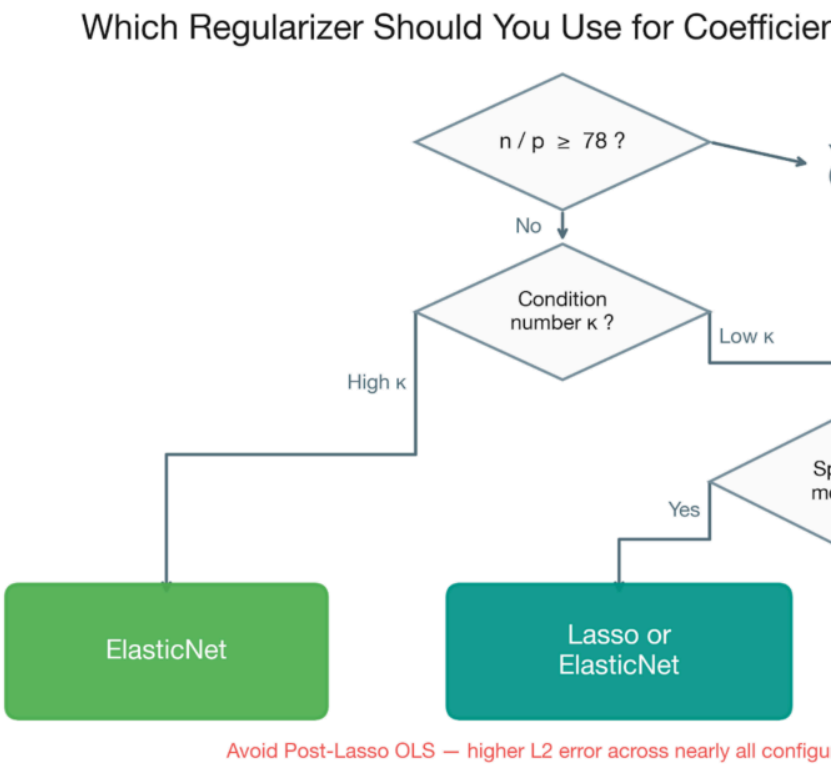

Figure 6: When the goal is coefficient estimation, Ridge becomes

Bottom line for coefficient estimation: Elastic
domain-dependent at low κ, never Post-Lasso

A Practitioner's Decision Guide

All of the findings above distill into a decision
branches exclusively on quantities you can co
fitting a single model: the sample-to-feature r
condition number κ (via numpy.linalg.cond(X)),

discrimination is needed, the regularization strength α elected by a quick LassoCV run as a proxy for the latent SNR.

The full flowchart is available in our paper (Figure 7). Here, we walk through the logic as a decision tree.

The under-determined regime

If your feature count exceeds your sample size under-determined regime. Lasso's α frequently upper boundary of the search grid here, and it **Default to Ridge or ElasticNet** for all objective with caution.

The large-sample regime

If n/p ≥ 78, you're in the large-sample regime converge. Performance gaps vanish across pre selection, and coefficient estimation simultan

Use RidgeCV. It's the fastest method by a wide is no accuracy penalty. If you specifically need for interpretability, ElasticNetCV or LassoCV ar this ratio. The choice among them is immateri

The regime where choice matters

Below *n/p* = 78 is where method choice matte regularizer depends on what you're optimizing

If prediction is your priority: Use RidgeCV. The among the core three methods are too small t complexity or compute. One narrow exception gh SNR (~1.0), ElasticNet offers a detectable

regardless of κ; at $n \approx 100$ with very low SNR, Ridge is marginally preferred. In either case, the margin is modest relative to the improvement available from increasing sample size.

If variable selection is your priority: Branch on the condition number.

- **$\kappa > \sim 10^4$** (high multicollinearity): Use ElasticNetCV. This is among the strongest recommendations in nuance: at moderate-to-high SNR (or $n \geq 1$ is clearly preferred, with F1 advantages ov ΔF1 of +0.75. At very low SNR with $n \approx 100$ saturated CV-elected α), Ridge achieves th only through perfect recall (retaining all fe genuine variable selection. If you need an model even in this corner, ElasticNet rema option and still vastly outperforms Lasso.
- **$\kappa < \sim 10^2$** (well-conditioned): An important **not default to Lasso even at low κ.** Lasso's sharply at lower SNR levels regardless of r while ElasticNet maintains recall ≥ 0.91 ac levels. ElasticNet is the safe default here. run a quick LassoCV and inspect the elect saturated at the boundary, you're in a low- provides the best F1 (though not through g sparsification). If α is moderate, stick with low *and* domain expertise suggests sparsit viable.

If coefficient estimation is your priority: Branc number.

- **$\kappa > \sim 10^4$:** ElasticNetCV dominates regardle

- **κ < ~10²:** Use domain knowledge. Sparse model → Lasso. Dense model → Ridge.

The α Diagnostic: A Free SNR Proxy

The one latent parameter that matters for fine-grained decisions, signal-to-noise ratio, can be approximated at zero additional cost. When scikit-learn's LassoCV fits your data, it reports the elected α. This value is inversely re underlying SNR: high α signals weak signal, low signal.

Our simulations provide direct empirical confir highest elected α values (approaching 10^4–10^5) exclusively in small-*n*, low-SNR configurations

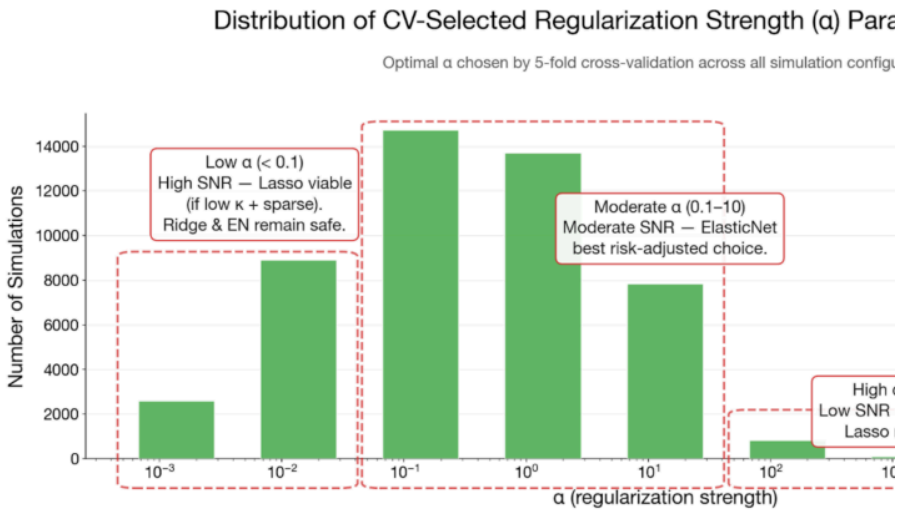

Figure 7: The regularization parameter α can be a useful pr

These thresholds are approximate heuristics d simulation grid, they'll vary with feature scalin characteristics. Treat them as guidelines, not s

In All Uncertain Cases

When you're unsure about SNR, unsure about sparsity, or operating in the intermediate-κ range we didn't directly test: **ElasticNet is the default that won't burn you**, and **Post-Lasso OLS should be avoided**.

The Meta-Finding: Sample Size Trumps Everything

One takeaway matters more than any method- **increasing your sample-to-feature ratio does objective than any regularizer choice.**

Sample size is the dominant driver of perform across all three metrics ($\omega^2 = 0.308$ for F1, a *la* × SNR interaction is the strongest two-way int comparisons ($F = 569$, $p < 0.001$). Signal-to-no precisely when samples are scarce. And at *n/p* choice becomes irrelevant entirely.

If you're spending days tuning your regularizer be growing your training set, you're optimizing

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Quick Reference

Objective	Default	When to Switch
Prediction	RidgeCV	ElasticNet only if $n \approx 100$ + high SNR (marginal gain)
Variable Selection	ElasticNetCV	Lasso only if: low κ + high SNR + sparse domain
Coef. Estimation	ElasticNetCV (high κ) / Ridge (low κ, dense)	Lasso at low κ + sparse

Note: When $n/p \geq 78$, all methods perform equivalently. Default to RidgeCV (fastes

Table 3: The most appropriate regularizer is determined by both the nature the research objective.

Putting It Into Practice

The simulation framework is a reusable harness. We capped sample sizes at 100k observations for compute reasons, but the grid still spans the n/p inflection point where regularizer performance shifts. We're extending it now to newer regularizers (Adaptive Lasso, SCAD, MCP) and intermediate κ levels.

To apply this framework to your next project, compute three quantities before you fit anything: the sample-to-feature ratio (n/p), the condition number (κ), and if you're ir regime, a quick LassoCV α as your SNR proxy. I decision guide above based on your primary ol

If n/p ≥ 78, use Ridge and spend your tuning b If $n/p < 78$ and κ is high, use ElasticNet and dc it. The only scenario where the choice requires low κ with small n, and even there, ElasticNet answer.

The full paper, including all appendix figures, A the consolidated decision flowchart, is availab

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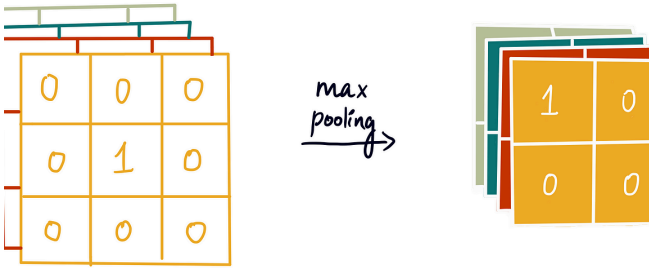

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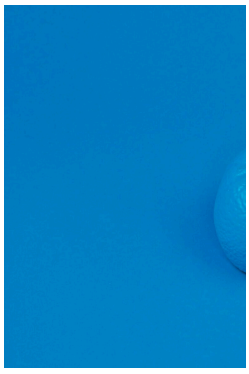

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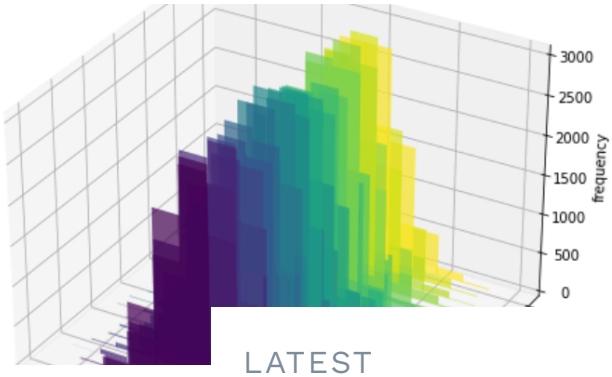

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